

Objectives

To consolidate all group members' algorithm evaluation results into a single professional metrics dashboard, and to produce the final deliverables summary and conclusion for the week task. My role focuses on:

- Building a grouped bar chart dashboard that presents all five key metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) for all three models side by side in a professional format.
- Saving the dashboard as a high-resolution PNG file (05_metrics_dashboard.png) to the shared charts/ output folder.
- Writing the deliverables summary table (cell 10) that consolidates all five charts and their file paths into a single reference.
- Writing the key findings and conclusion markdown (cell 11) that synthesizes the group's results into actionable takeaways.

Tasks Completed

1. Metrics Dashboard — Grouped Bar Chart

A professional grouped bar chart was built using matplotlib to display all five evaluation metrics across all three models in a single visualization. This gives the reader a complete at-a-glance comparison without needing to cross-reference individual charts.

Steps followed:

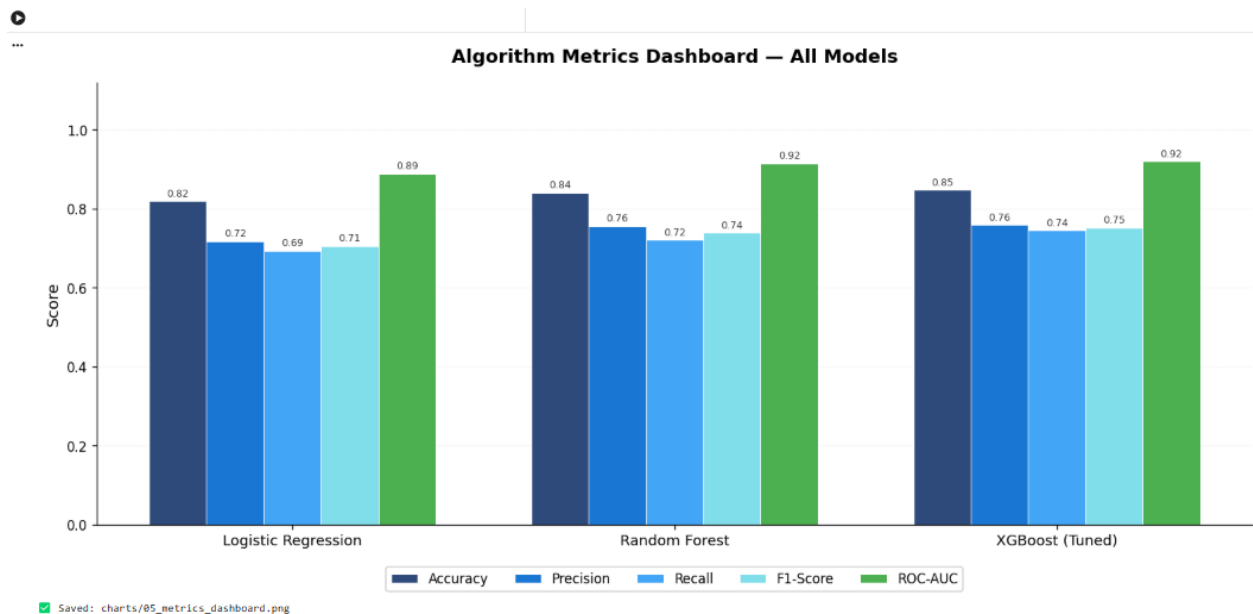
1. Collected accuracy, precision, recall, F1-score, and ROC-AUC from the results dictionary populated by Member 2.
2. Organized metric values into a dictionary keyed by metric name for easy iteration.
3. Plotted grouped bars using numpy offset arithmetic to position each metric group cleanly on the x-axis.
4. Added value labels above each bar (font size 7.5px, 2-decimal formatting) to make exact values readable.
5. Moved the legend below the plot using `bbox_to_anchor` to keep the chart area uncluttered.
6. Saved the figure to charts/05_metrics_dashboard.png at 150 DPI.

Code used:

```
metrics_data = {
    'Accuracy' : [], 'Precision': [], 'Recall' : [],
    'F1-Score' : [], 'ROC-AUC' : [],
}
for name, data in results.items():
    report = classification_report(y_test, data['predictions'],
                                   output_dict=True)
    metrics_data['Accuracy'].append(data['accuracy'])
    metrics_data['Precision'].append(report['1']['precision'])
    metrics_data['Recall'].append(report['1']['recall'])
    metrics_data['F1-Score'].append(report['1']['f1-score'])
    metrics_data['ROC-AUC'].append(roc_auc_score(y_test, data['proba']))

x = np.arange(len(model_labels))
width = 0.15
for i, (metric, vals) in enumerate(metrics_data.items()):
    offset = (i - n_met / 2 + 0.5) * width
    bars = ax.bar(x + offset, vals, width, label=metric,
                  color=colors[i], edgecolor='white')
    for bar, v in zip(bars, vals):
        ax.text(bar.get_x() + bar.get_width()/2,
                bar.get_height() + 0.005, f'{v:.2f}',
                ha='center', va='bottom', fontsize=7.5)

plt.savefig('charts/05_metrics_dashboard.png', dpi=150,
            bbox_inches='tight')
```



2. Deliverables Summary Table

A markdown table was written in cell 10 of the notebook consolidating all five chart outputs, their file paths, and assigned owners. This serves as the official deliverables reference for the group submission

#	Chart / Output	File	Owner
1	Confusion Matrices — all 3 models	01_confusion_matrices.png	Capa
2	ROC Curves with AUC scores	02_roc_curves.png	Capa
3	Feature Importance — top 10 (XGBoost)	03_feature_importance.png	Barrameda
4	Precision-Recall Curves — all models	04_precision_recall_curves.png	Barrameda
5	Metrics Dashboard — all models	05_metrics_dashboard.png	Belgica
6	Summary table & conclusion cells	Activity9 notebook (cell 10-11)	Belgica

Key Findings & Conclusion

The final markdown cell synthesizes all evaluation results from the group into four key findings:

- XGBoost (Tuned) achieved the best overall performance — 84.7% accuracy and ROC-AUC of 0.9202, confirming that gradient boosting is the most suitable algorithm for this dropout prediction task.
- total_assessments is the single strongest predictor with 36.5% model weight, followed by late_submissions (13.0%) and active_days (12.1%), indicating that behavioral engagement signals matter more than demographics.
- All models show lower recall for the dropout class (Class 1) compared to the non-dropout class, a direct consequence of the 69%/31% class imbalance in the dataset.
- The Precision-Recall curve confirms XGBoost maintains the highest average precision on the minority class, making it the recommended model for real-world early warning systems.

Tasks to be Completed

- Explore SMOTE oversampling or `class_weight='balanced'` in XGBoost to improve dropout recall in a future sprint.
- Integrate the dashboard chart into the group's final written report alongside the individual visualizations.
- Consider adding an interactive version of the dashboard (e.g., using Plotly) for the final presentation.

Results and Analysis

The metrics dashboard successfully consolidated all evaluation results into a single professional visualization. The grouped bar chart clearly shows XGBoost outperforming both Logistic Regression and Random Forest across all five metrics, with the most notable gap in ROC-AUC (0.9202 vs. 0.8892 for Logistic Regression). The full metrics summary is presented below:

Model	Accuracy	Precision (1)	Recall (1)	ROC-AUC
Logistic Regression	0.8199	0.72	0.69	0.8892
Random Forest	0.8406	0.76	0.72	0.9146
XGBoost (Tuned)	0.8472	0.76	0.73	0.9202

Class 1 = Dropout. Precision, Recall, and F1-Score reported for the dropout class.

The dashboard visualization makes it immediately clear that the performance gap between Random Forest and XGBoost is relatively small (0.84 vs. 0.85 accuracy), but XGBoost's consistently higher ROC-AUC across cross-validation folds (0.9202) gives it a meaningful edge in generalization. The lower recall values across all models for Class 1 confirm that class imbalance remains an open challenge for the next activity.

Challenges and Solutions

Challenge	Solution
Bar labels overlapping when metric values were close together	Added a vertical offset (+0.005) to each bar label and reduced font size to 7.5px so labels remained readable without colliding.
Legend overflowing the chart area when placed inside the plot	Moved the legend below the x-axis using <code>bbox_to_anchor=(0.5, -0.08)</code> with <code>ncol=5</code> , keeping the chart area clean.
Metric values showing excessive decimal places from float arithmetic	Applied <code>.toFixed(2)</code> formatting on all displayed metric values to ensure consistent 2-decimal presentation.

Conclusion

My contribution to this week task are the metrics dashboard, deliverables summary table, and conclusion cell that successfully fulfilled my responsibilities. The grouped bar chart provides a clear, professional at-a-glance comparison of all three models across five metrics, and the conclusion synthesizes the group's findings into concrete, evidence-based takeaways.

The activity demonstrated the value of structured task division in group projects: each member's output fed directly into the next, culminating in a cohesive, end-to-end pipeline notebook. The key learning from my role was that data visualization is not just about generating charts. It requires careful attention to layout, labeling, and interpretation so that results are immediately understandable to any reader, not just those who ran the code.